Group Assignment: Noise Schedule Ablation in Diffusion Models

Foundation Models and Generative AI

Overview

The noise schedule is a core design element of diffusion models. It controls how information is progressively corrupted during the forward process and how the model learns to reverse that corruption. This assignment studies what happens when a pretrained image-to-image diffusion model is fine-tuned on a new dataset under a power-transformed version of its native noise schedule. Each group selects a pretrained model, a specific image-to-image task, and a dataset, fine-tunes the model under two schedules, and analyzes the results across three conditions.

Phase 1 — Declaration

Submit by October 10, as a PDF on Google Classroom. The declaration must contain:

- The image-to-image task the group has chosen, stated precisely. Examples include toonification of real face photographs, sketch-to-photorealistic image synthesis, low-resolution to high-resolution image synthesis, daytime-to-nighttime scene conversion, and grayscale-to-color image translation. These are examples — any clearly defined image-to-image translation task is acceptable.
- Link to the paper or codebase of the chosen pretrained model
- Dataset name and source URL
- Evidence that the dataset was not used during pretraining of the chosen model — either (a) a link to the paper or codebase, with page number and line number, confirming the dataset is absent from the pretraining data, or (b) a link to the paper or codebase, with page number and line number, describing the pretraining data domain, and a one-paragraph argument that the chosen dataset belongs to a clearly different domain
- The native noise parameterization of the chosen model — state which quantity serves as θ_t (for example, β_t, α_t, or $\bar{\alpha}_t$) and confirm that $\theta_t \in (0, 1)$ for all t
- `Chosen exponent x, to two decimal places, where x` $\in$ `(1.5, 3.0)`

The Experiments

Different diffusion models parameterize the forward noise process differently. Let θ_t denote the native noise parameter of the chosen pretrained model at timestep t, where t = 1, 2, ..., T and T is the total number of diffusion timesteps. Depending on the model, θ_t may be β_t, α_t, $\bar{\alpha}_t$, or another scheduler-specific quantity. The group must identify which parameterization their model uses and state it explicitly.

The power transformation applied in this assignment is only well-defined and interpretable when $\theta_t \in (0, 1)$ for all t. Groups must verify and confirm this in their declaration and in the Schedule Analysis section of the report. Models parameterized by quantities that can exceed 1 are not suitable for this

assignment unless the group converts to an equivalent bounded parameterization and justifies the conversion.

Each group runs three experiments:

- **Run 0** — apply the pretrained model directly to the group's input images with no fine-tuning; save the outputs as the pre-fine-tuning baseline
- **Run A** — fine-tune the pretrained model on the chosen dataset using the original schedule $\{\theta_t\}$
- **Run B** — fine-tune the pretrained model on the chosen dataset using the modified schedule $\{\theta_t^x\}$, where x is the group's declared exponent, applied elementwise

All hyperparameters, including optimizer, learning rate, batch size, number of epochs, dataset preparation, and train/validation/test split, must be identical across Run A and Run B. Use the same pretrained checkpoint for both fine-tuning runs. The only difference between Run A and Run B is the noise schedule.

Phase 2 - Submission Components

1. Code

- The modified codebase with every changed line marked # MODIFIED and every added line marked # ADDED
- The exponent x declared as a named constant at the top of the fine-tuning script
- A README specifying the exact commands to reproduce Run 0, Run A, and Run B, the random seed used, and the software environment (Python version, library versions)

2. Fine-tuning Logs

- One CSV file per fine-tuning run (Run A and Run B) with columns: epoch, train_loss
- At 5 equally spaced fine-tuning epochs, at least 3 generated samples per checkpoint for both Run A and Run B, clearly labeled with run name and epoch number

3. Inference Trajectories

Using the final fine-tuned model from Run A and Run B, save the noisy sample x_t at 5 equally spaced diffusion timesteps during the reverse process, for at least 3 samples per timestep, clearly labeled with run name and timestep index.

4. Report

The report must contain the following sections in order:

Task and Model

A precise description of the chosen image-to-image task. A brief description of the chosen pretrained model and its forward process. Identify and define the native noise parameterization θ_t. All symbols defined. Link to the baseline paper or codebase.

Dataset and Verification

Description of the chosen dataset — number of samples, image resolution, and why it is appropriate for the chosen task. Evidence that the dataset was not used during pretraining, with link, page number, and line number as described in the declaration.

Schedule Analysis

Confirm that $\bar{\alpha}_t \in (0, 1)$ for all t, with justification. Plot $\{\bar{\alpha}_t\}$ and $\{\bar{\alpha}_t^x\}$ against t on the same axes. Describe analytically what changes in the forward process when $x > 1$. Explain how the signal-to-noise ratio at each timestep is affected by the exponent.

Experimental Setup

- Dataset preparation procedure: how the dataset was collected, cleaned, and preprocessed
- Train, validation, and test split: sizes and how the split was performed
- Fine-tuning hyperparameters: optimizer, learning rate, batch size, number of epochs, and any other relevant details; these are identical across Run A and Run B
- How validation was performed during fine-tuning: metric used and frequency
- How the test set was used for final evaluation

Implementation

- Which file(s) in the codebase were modified and which lines changed
- The original lines and the modified lines shown side by side, clearly labeled
- How the group verified that Run B was using $\{\bar{\alpha}_t^x\}$ and not $\{\bar{\alpha}_t\}$

Results

- Output samples from Run 0 on the group's input images
- Fine-tuning loss curves for Run A and Run B on a single plot
- Generated samples at each of the 5 equally spaced fine-tuning checkpoints for Run A and Run B, clearly labeled
- Inference trajectory images (noisy sample x_t) at 5 equally spaced diffusion timesteps for Run A and Run B, clearly labeled
- A table of test-set scores for Run 0, Run A, and Run B on the chosen metric; the chosen metric must be justified in one sentence

Analysis

Which fine-tuning run produced better outputs and at which checkpoint the difference first became visible. A mechanistic explanation of the observed result grounded in the schedule mathematics. An explanation of how Run 0 outputs compare to Run A and Run B, and what this reveals about the effect of fine-tuning on an out-of-distribution dataset.

Takeaway

What the group observed and learned from running this experiment.

Contribution Statement

Who was responsible for which component, signed by all group members.